

Mnl theory assignment 3

Let (X, ϵ) be a measureable space and consider a function $f: X \rightarrow \mathbb{R}$

a) prove the validity of the implication,

$$f \in \mathcal{M}(\epsilon) \Rightarrow |f| \in \mathcal{M}(\epsilon)$$

Also stated:

$$f \text{ is } \epsilon - \mathbb{B}(\mathbb{R}) \text{ measureable} \Rightarrow |f| \text{ is } \epsilon - \mathbb{B}(\mathbb{R}) \text{ measureable}$$

We know that $f \in \mathcal{M}(\epsilon) \Leftrightarrow -f \in \mathcal{M}(\epsilon)$

We define 2 sets and their inhereted sigma alegbras:s

$$A_+ = \{x \in X: f(x) \geq 0\}, \quad \epsilon_{A_+} = \{i_{A_+}^{-1}(B): B \in \epsilon\} = \{A_+ \cap B: B \in \epsilon\}$$

$$A_- = \{x \in X: f(x) < 0\}, \quad \epsilon_{A_-} = \{i_{A_-}^{-1}(B): B \in \epsilon\} = \{A_- \cap B: B \in \epsilon\}$$

And we need to show that the for the following funktions

$$f_1: A_+ \rightarrow \mathbb{R}$$

$$f_2: A_- \rightarrow \mathbb{R}$$

$$f_1 \in \mathcal{M}(\epsilon_{A_+}), f_2 \in \mathcal{M}(\epsilon_{A_-})$$

Well start with f_1

We must show that

$$f_1^{-1}((-\infty, b]) = \{x \in A_+: f_1(x) \leq b\} \in \epsilon_{A_+}$$

But

$$f_1^{-1}((-\infty, b]) = \{x \in A_+: f(x) \leq b\} = f^{-1}((-\infty, b]) \cap A_+$$

And since f is measureable $f^{-1}((-\infty, b]) \in \epsilon \Rightarrow f^{-1}((-\infty, b]) \cap A_+ \in \epsilon_{A_+}$

And thus $f_1 \in \mathcal{M}(\epsilon_{A_+})$

Analogously for f_2

Thus $f_2 \in \mathcal{M}(\epsilon_{A_-})$ and by extension $-f_2 \in \mathcal{M}(\epsilon_{A_-})$

Then by construction $|f|: X \rightarrow \mathbb{R}$

$$|f|(x) = \begin{cases} f_1(x) & \text{for } x \in A_+ \\ -f_2(x) & \text{for } x \in A_- \end{cases}$$

Since A_+, A_- are a partition of X

$$f_1 \in \mathcal{M}(\epsilon_{A_+}), -f_2 \in \mathcal{M}(\epsilon_{A_-})$$

Then by the tuborg rule we must have $|f| \in \mathcal{M}(\epsilon)$

b) just need a counter example

let $X = [0,1]$

we define: $f: X \rightarrow \mathbb{R}$

$$f(x) = \begin{cases} 1 & \text{for } x \in A \\ -1 & \text{for } x \in A^c \end{cases}$$

Where $A \subset X$ is a non measureable set

$$|f|(x) = 1$$

Is a constant function and thus measureable

Since $f^{-1}(\{1\}) = X \in \epsilon$

$$f^{-1}(\{1\}) = A$$

Is non measurable thus f is non measurable